

ELECTRONIC STORE MANAGER

FIRST PROJECT REPORT

1. Introduction

Electronic Store Manager is a software that simulates an electronic retail environment by addressing needs such as: maintaining data about all items in the store, tracking sold items along with total income and expenses, making each role type be able perform their specified operations and overseeing each employee in the store. The program will be able to provide an interactive interface for every employee in the company, with each program instance providing the specific views and functionalities of the user that is logged in to operate the software to their needs.

2. Work Plan

- Endri Baku (Team Leader) - System Design, overseeing of the development phases, Backend Developer, File Handling
- Moel Plasa (Member) - Backend Developer, File Handling
- Hazis Voda (Member) - UI Developer, Testing and Debugging
- Daron Delvina (Member) - Input and Validation, Exception Handling

3. Overall Timeline of the Project

So, the development of the project as agreed to by the group members is split into these phases:

Phase 1: Requirements process (research, specification, validation) and Design (UML Diagrams)

Deadline: 27 December

Phase 2: Prototype of System with UI and Backend Functionality

Deadline: 10 January

Phase 4: Testing and Debugging

Deadline: To be Decided (depends on the date of the announced project deadline)

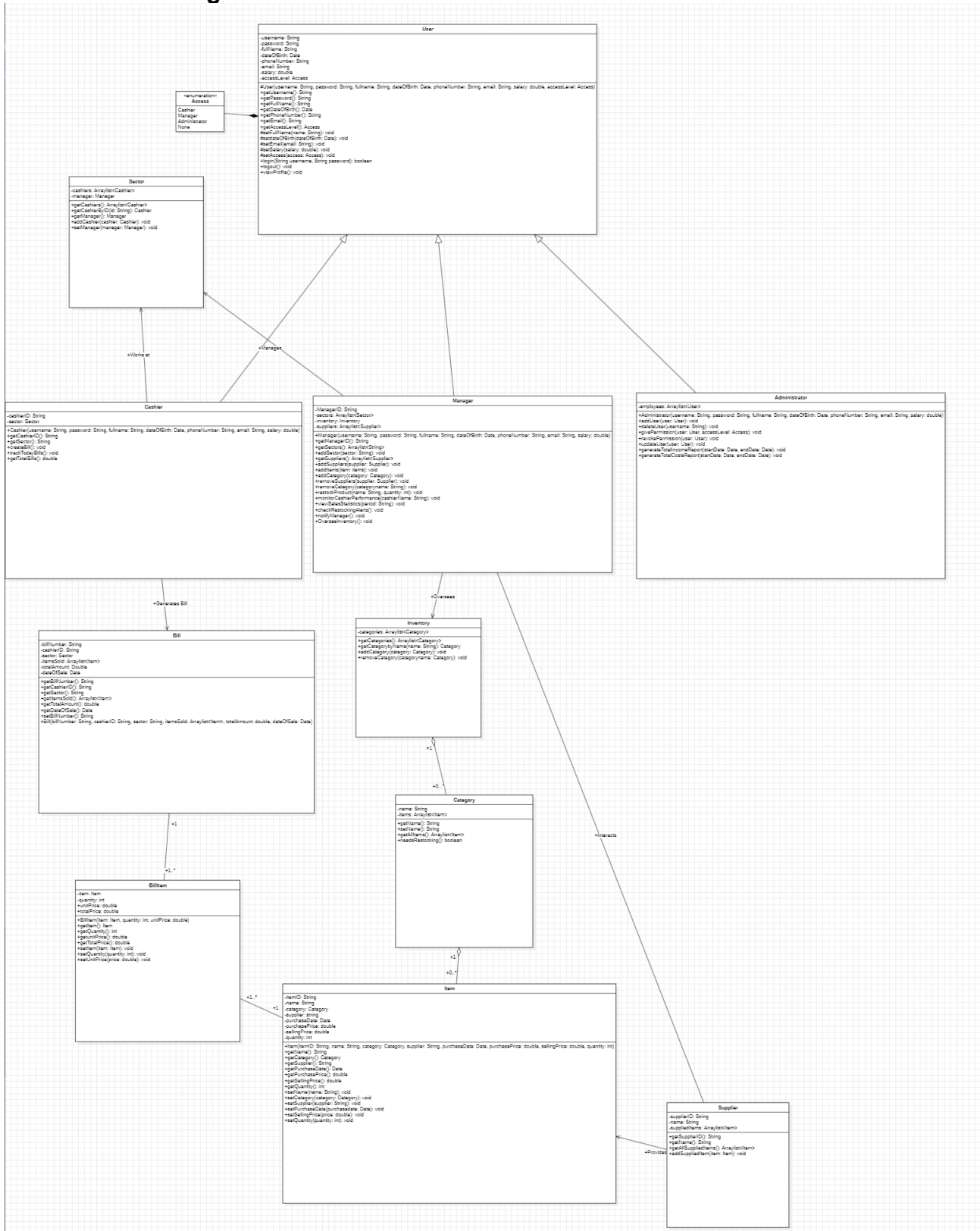
Phase 5: Finalization

Deadline: To be Decided (depends on the date of the announced project deadline)

4. Planned Design of the Project

The core design of the project (while small changes/tweaks can be made during the implementation and development process) has been curated by the Team Leader and agreed upon all the other group members. The project will follow the Model-View-Controller approach as specified in the project requirements. The design of the project has been made using UML Diagrams shown below:

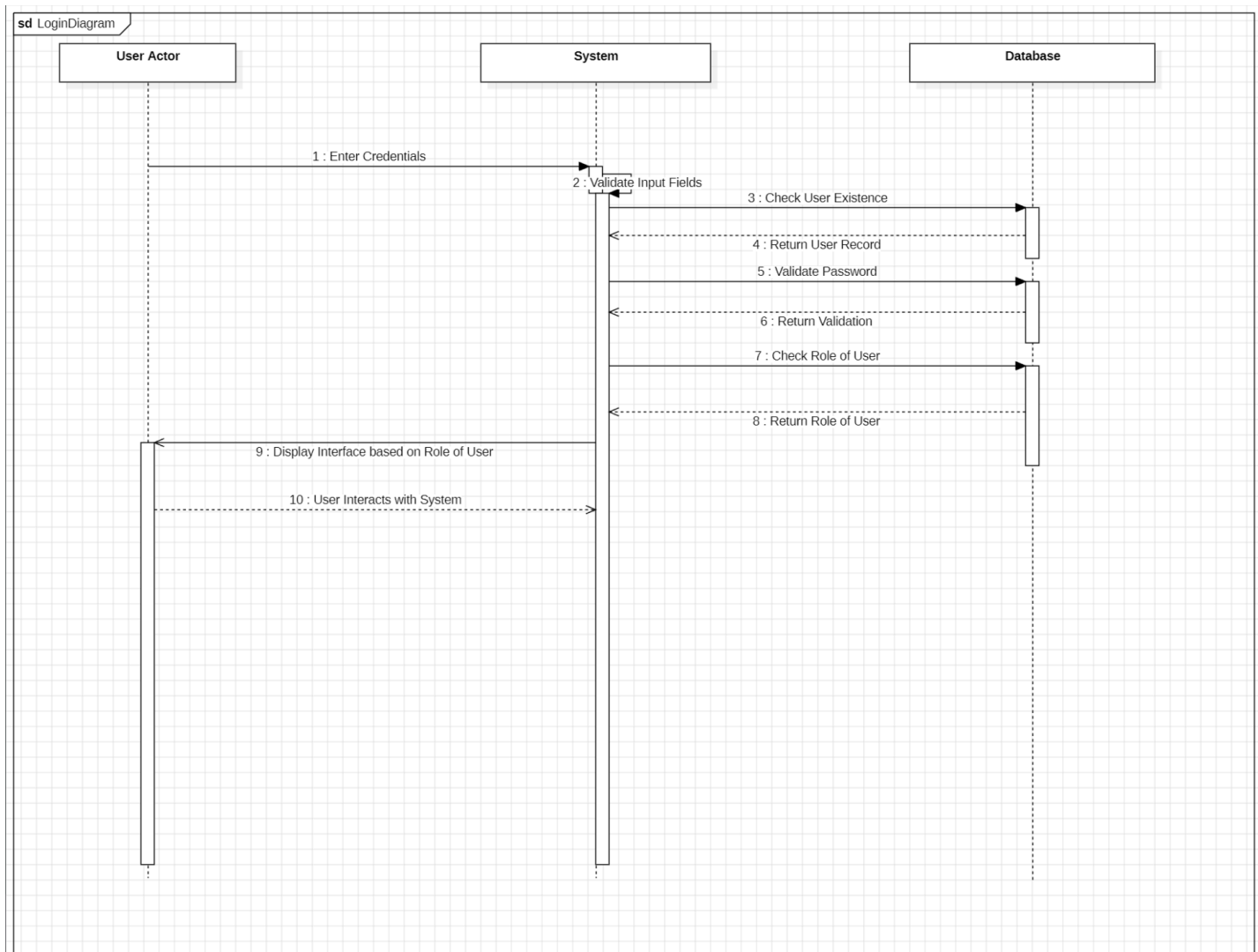
1. Class Diagram of each core element/class of the software

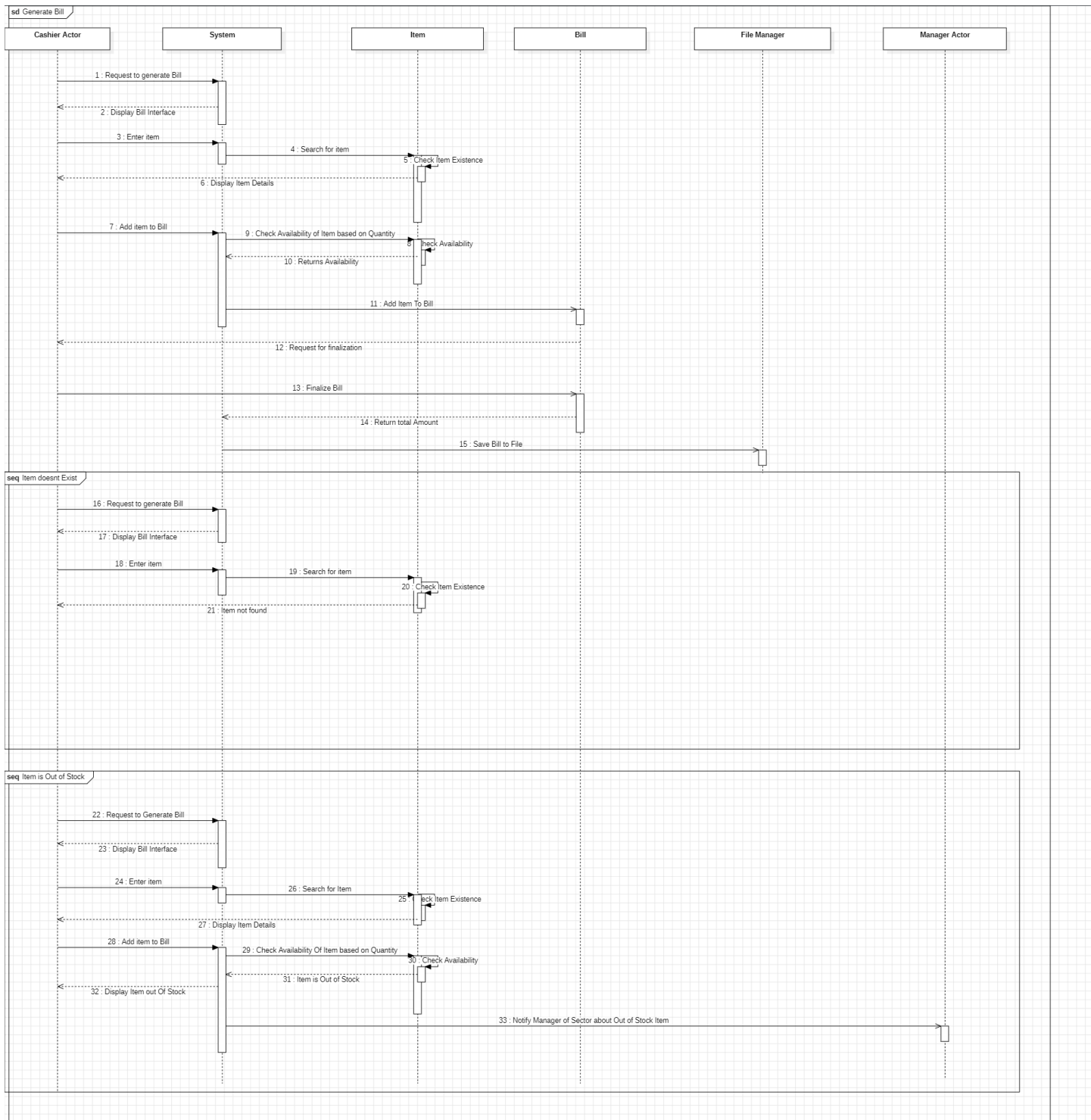


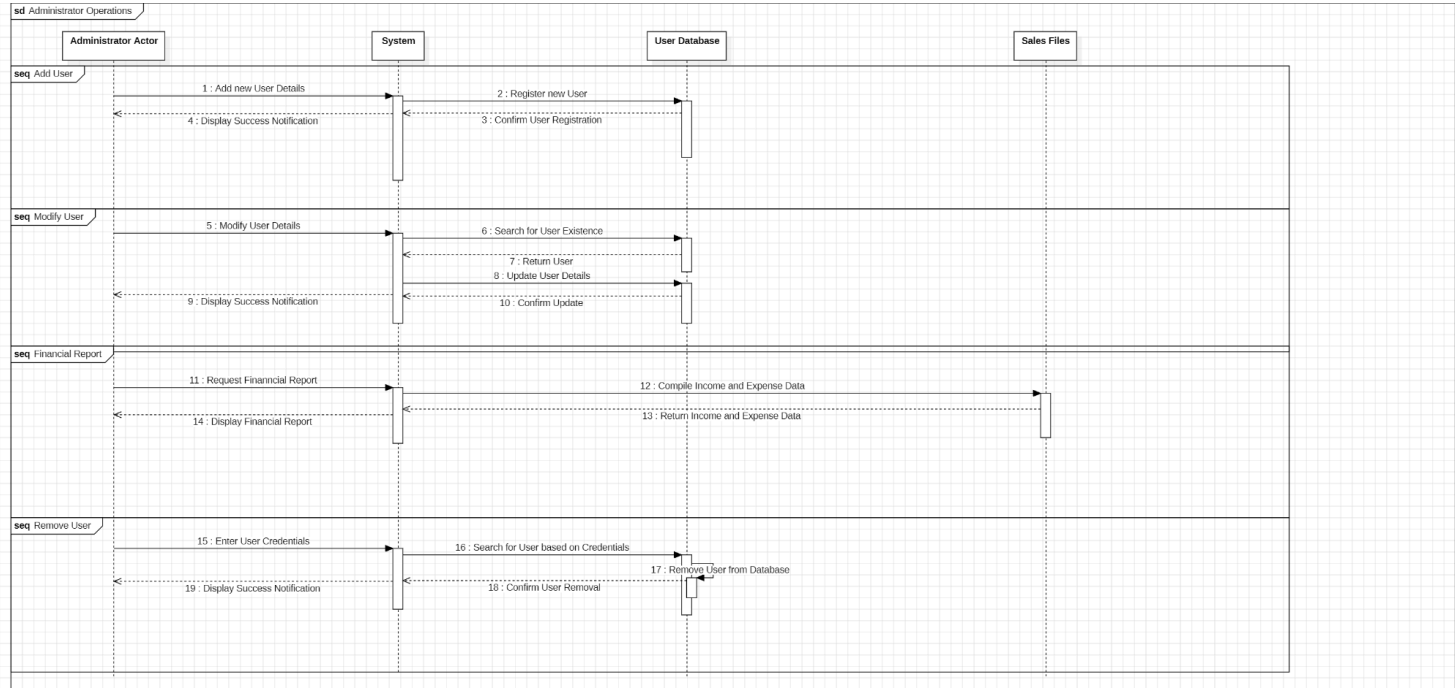
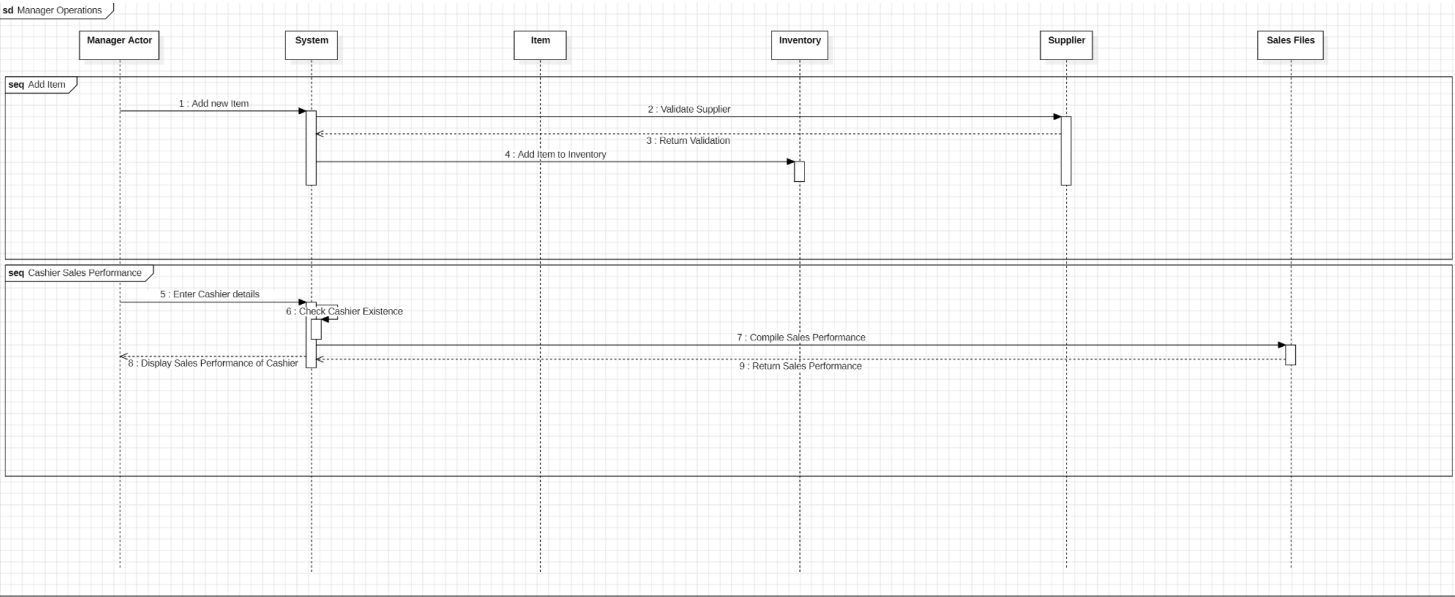
Notes:

- Each class that will contain data that is going to be saved using files will also implement the **Serializable** Interface (Not shown in this diagram)
- Each class can be subject to changes in the future (maybe in method implementation) but the changes have to be very small and not detrimental to the whole structure of the system.

2. Sequence Diagram



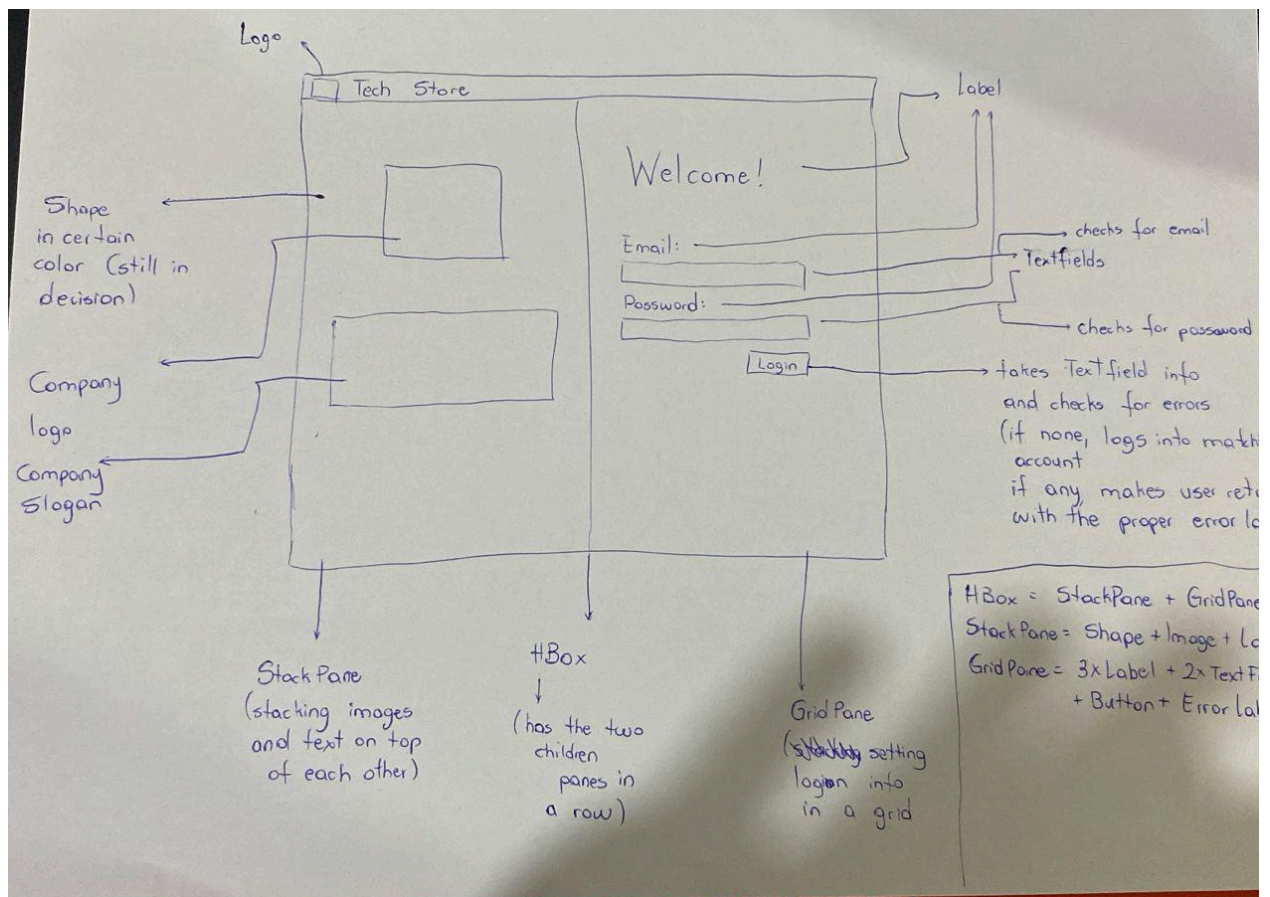




Notes:

- These diagrams show various situations of how elements of the system would interact with each other (what messages are sent as when), also demonstrated with alternative sequences.
- These sequences are subject to small changes when the implementation phase is ongoing.

3. Early Concept/Prototype of the Interface



5. Possible Challenges

During the development phase, it is possible that we are going to run into some issues regarding things that we can say we are more unfamiliar of (file I/O, JavaFX User Interface) as we believe that those would be the most challenging parts of the implementation.

Our approach to dealing with these possible issues is a careful and controlled development phase, with each developer of a specific part of the program communicating with developers of other parts of the project to ensure that development commitments of one part of the project won't create issues for others.

Development will consist of frequent commitments through GitHub and progress will be tracked through Trello.

6. Recap and Next Steps

For this first project report, we included our work plan, planned deadlines, our designs and we pointed out some possible challenges that we could face during implementation. Through our diagrams we showed designs that adhered to the Object-Oriented programming approach and MVC Architecture. Our next steps are finalizing the system design and beginning implementation.

We are committed to show an application that is user-friendly, fully functional and concise.